Practice workbook · Explain how ADH changes urine volume and concentration

Hypothalamus detection, pituitary release of ADH, collecting duct permeability, and explaining urine differences on hot days, after drinking water or alcohol, or between samples.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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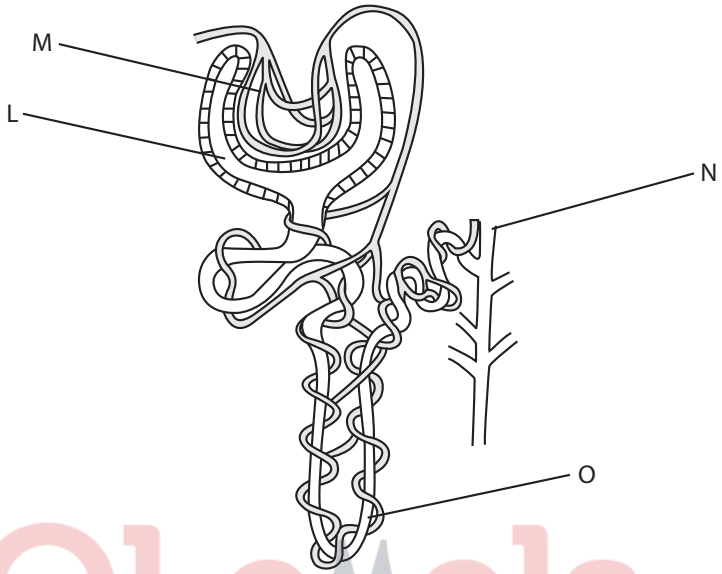
Type ID: HBI-ADH-OSMOREGULATION

1. Question 6 · part c · 5 marks

November 2023 Paper 1

Worked-example retry · Attempt part c only.

6 The diagram shows a kidney tubule.



(a) Name structures L, M, N, and O.

(4)

L .....

M .....

N .....

O .....

(b) (i) Describe the process that causes fluid to be formed in L.

(3)

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1. Question 6 · part c · 5 marks

November 2023 Paper 1

Worked-example retry · Attempt part c only.

(ii) Name **two** components of the blood that do not pass into L.

(2)

1 .....

2 .....

(iii) Label, with the letter G, the part of the kidney tubule where most of the glucose is reabsorbed.

(1)

(c) Explain why the volume of urine a person produces on a hot day is less than the volume of urine the same person produces on a cold day.

(5)

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(Total for Question 6 = 15 marks)

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1. Question 6 · part c · 5 marks · Working and answer

Working and answer

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2. Question 7 · part a.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part a.i only.

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7 The renal system is involved in osmoregulation.

(a) (i) Name the area of the brain that detects the water content of the blood.

(1)

.....

(ii) The kidneys are part of the renal system.

Put one tick (✓) in each column of the table to identify the function of different structures in the kidney.

(3)

	Structure		
Function	**glomerulus**	**proximal tubule**	**renal vein**
regulation of water level			
ultrafiltration			
reabsorption of glucose			
removal of blood from kidney			
supply of blood to kidney			
removal of urine from collecting duct			

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2. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

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3. Question 6 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The effects of alcohol

Humans have been drinking alcohol for thousands of years. Drinking too much alcohol over many years can cause diabetes, malnutrition and diseases of the central nervous system and the liver. One immediate side effect of drinking too much alcohol is increased urination.

Beer is about 95% water and only 5% alcohol. The liver converts that 5% of alcohol into the same volume of water and some carbon dioxide. So if a person drinks 200 cm^3 of beer (one unit), the end result is 200 cm^3 of water. However, this person will not just urinate 200 cm^3 of urine, in fact the person urinates a total of 320 cm^3 of urine. So in general each unit of alcohol makes the person urinate an extra 120 cm^3 of urine more than the normal urine output.

A person with a mass of 60 kg produces 60 cm^3 of urine an hour and a person with a mass of 80 kg produces 80 cm^3 an hour and so on. Therefore if a 60 kg person drinks 200 cm^3 of beer in an hour they will produce 60 cm^3 of urine plus 320 cm^3, making a total of 380 cm^3.

Alcohol interferes with the mechanism that regulates water levels in our body because alcohol affects a small gland called the pituitary gland. One hormone released by the pituitary gland is anti-diuretic hormone (ADH). Alcohol reduces how much ADH is released by the pituitary. Each unit of alcohol that a person drinks forces the kidneys to produce an extra 120 cm^3 of urine more than the normal output.

Logic suggests that drinking lots of water would replace the 120 cm^3 of urine produced, but it is not that simple. The body only retains half the water that a person drinks. The other half leaves the body in urine, so the person will become dehydrated even if they drink lots of water.

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3. Question 6 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

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(a) (i) Where in the body is the pituitary gland (line 16)?

(1)

- ☒ **A** brain
- ☒ **B** kidney
- ☒ **C** liver
- ☒ **D** pancreas

(ii) Which system includes the pituitary gland (line 16)?

(1)

- ☒ **A** digestive
- ☒ **B** endocrine
- ☒ **C** nervous
- ☒ **D** skeletal

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(b) A person weighs 75 kg and drinks 400 cm^3 of beer in an hour (lines 11–14).

(i) Calculate the volume of urine produced by the person in this hour.

(3)

volume of urine = ... cm^3

(ii) Calculate the percentage increase in the volume of urine produced by this person in this hour.

(3)

percentage increase = ...%

3. Question 6 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

(c) Describe the effect of ADH on the kidney (line 17).

(2)

(d) Suggest the effects of dehydration on the blood after increased urination.

(2)

(e) Explain why half of the extra water that the person drinks still leaves the body in the urine (lines 21–24).

(3)

(Total for Question 6 = 15 marks)

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3. Question 6 · part c · 2 marks · Working and answer

Working and answer

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4. Question 5 · part c · 4 marks

June 2022 Paper 1

Further practice · Attempt part c only.

5 (a) There is a difference in the volume of urine produced on a hot day and the volume of urine produced on a cold day.

Design an investigation to determine the difference in the volume of urine produced on a hot day and the volume of urine produced on a cold day.

Include experimental details in your answer and write in full sentences.

(4)

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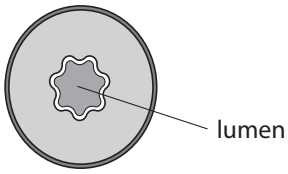
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4. Question 5 · part c · 4 marks

June 2022 Paper 1

Further practice · Attempt part c only.

(b) The diagram shows a cross-section of a blood vessel in the skin on a cold day.



Explain the changes in this blood vessel on a hot day.

(3)

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(c) Describe how ADH regulates the volume of water in the body on a hot day.

(4)

(Total for Question 5 = 11 marks)

4. Question 5 · part c · 4 marks · Working and answer

Working and answer

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5. Question 7 · part b.ii · 5 marks

January 2023 Paper 2

Further practice · Attempt part b.ii only.

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7 The renal system is involved in osmoregulation.

(a) (i) Name the area of the brain that detects the water content of the blood.

(1)

.....

(ii) The kidneys are part of the renal system.

Put one tick (✓) in each column of the table to identify the function of different structures in the kidney.

(3)

	Structure		
Function	**glomerulus**	**proximal tubule**	**renal vein**
regulation of water level			
ultrafiltration			
reabsorption of glucose			
removal of blood from kidney			
supply of blood to kidney			
removal of urine from collecting duct			

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5. Question 7 · part b.ii · 5 marks

January 2023 Paper 2

Further practice · Attempt part b.ii only.

(b) The table shows the percentage composition of two urine samples taken from one person.

Not all of the components are shown.

The first sample was taken at midday and the second sample was taken four hours later.

Component	Percentage composition	
	first sample	second sample
ammonia	0.05	0.08
uric acid	0.03	0.08
ions	1.34	1.46
urea	2.00	2.80
water	95.00	91.50

(i) Describe how urea is produced in the body.

(2)

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5. Question 7 · part b.ii · 5 marks

January 2023 Paper 2

Further practice · Attempt part b.ii only.

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(ii) The ADH response changes the percentage composition of urine.

Explain the ADH response that occurred in the person's body between the first and the second sample being collected.

(5)

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(Total for Question 7 = 11 marks)

5. Question 7 · part b.ii · 5 marks · Working and answer

Working and answer

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6. Question 9 · part whole · 5 marks

June 2024 Paper 1R

Further practice · Attempt part whole only.

9 Explain the role of ADH in osmoregulation.

(5)

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(Total for Question 9 = 5 marks)

6. Question 9 · part whole · 5 marks · Working and answer

Working and answer

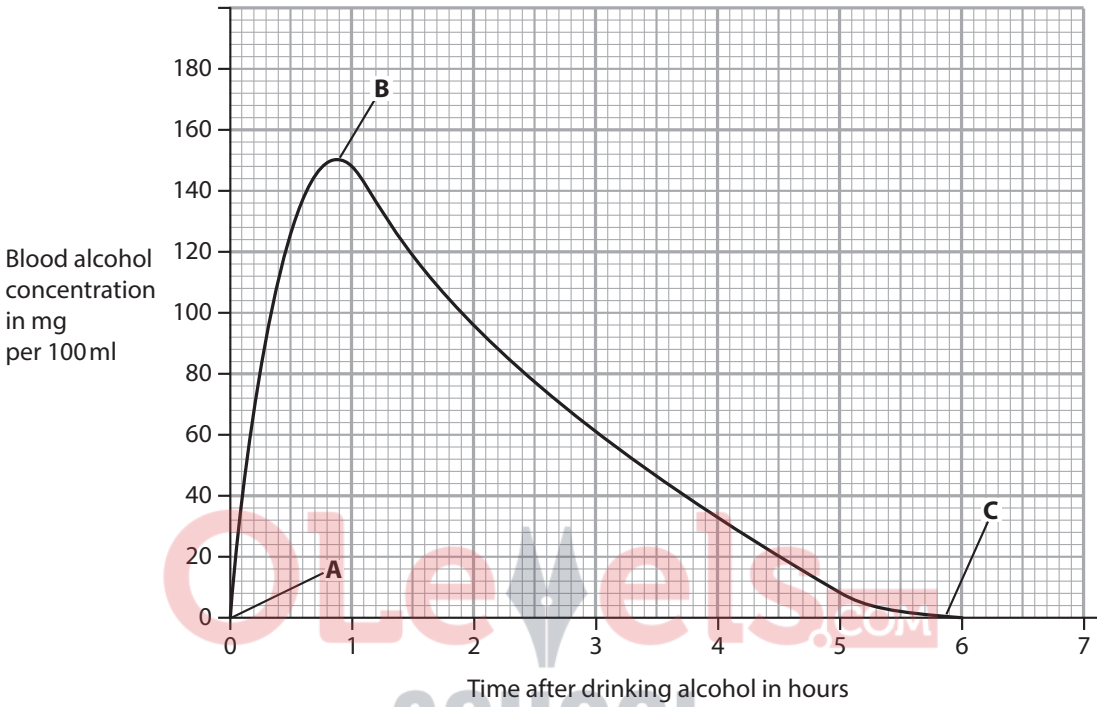
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7. Question 6 · part c · 5 marks

June 2025 Paper 1

Further practice · Attempt part c only.

6 The graph shows the changes in blood alcohol concentration of a person who drinks alcohol.



(a) (i) Explain the shape of the graph between points A and B.

(2)

(ii) Explain the shape of the graph between points B and C.

(3)

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7. Question 6 · part c · 5 marks

June 2025 Paper 1

Further practice · Attempt part c only.

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(b) In the United Kingdom the legal limit to drive is less than 80 mg per 100 ml of alcohol in the blood.

(i) The person shown in the graph drinks alcohol and the alcohol level in the blood goes above the legal limit to drive.

Determine how many hours after drinking alcohol it would be legal for the person shown in the graph to drive.

(1)

(ii) Explain why it is against the law to drive with high levels of alcohol in the blood.

(4)

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7. Question 6 · part c · 5 marks

June 2025 Paper 1

Further practice · Attempt part c only.

(c) Alcohol reduces the production of ADH.

Explain the effect on the urine produced by a person drinking alcohol.

(5)

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(Total for Question 6 = 15 marks)

7. Question 6 · part c · 5 marks · Working and answer

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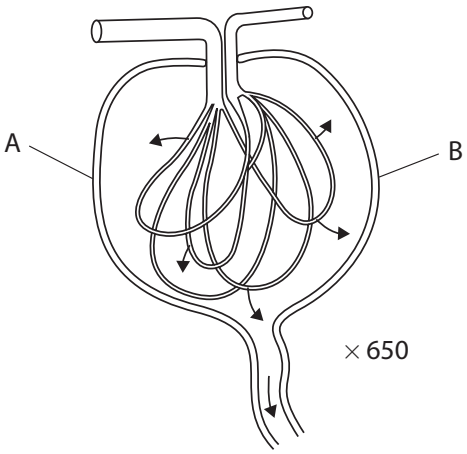
8. Question 1 · part b.iii · 3 marks

January 2022 Paper 2

Further practice · Attempt part b.iii only.

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1 (a) The diagram shows the Bowman's capsule and glomerulus of a kidney tubule (nephron).



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(i) Complete the diagram to show the missing parts of the kidney tubule.

(3)

(ii) Calculate the actual width of the Bowman's capsule, in mm, from the point labelled A to the point labelled B.

(3)

width of Bowman's capsule = ... mm

8. Question 1 · part b.iii · 3 marks

January 2022 Paper 2

Further practice · Attempt part b.iii only.

(iii) Name two components of the blood that would not pass into the Bowman's capsule.

(2)

1 .....

2 .....

(iv) Name one component of the glomerular filtrate that is completely reabsorbed into the blood from the tubule.

(1)

.....

(b) A scientist uses this method to investigate the effect of room temperature on urine production.

On day 1, a person drinks 1 dm^3 of water. Urine samples from the person are collected at intervals during a time period of four hours.

The room temperature is 20 °C.

This is then repeated on day 2, when the room temperature is 30 °C.

The table shows the results.

Time after drinking in hours	Volume of urine in cm^3	
	day 1 (20 °C)	day 2 (30 °C)
0.5	70	50
1.5	175	70
2.0	300	200
2.5	250	190
3.0	210	150
4.0	100	50

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8. Question 1 · part b.iii · 3 marks

January 2022 Paper 2

Further practice · Attempt part b.iii only.

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(i) State two factors that should be controlled in the investigation.

(2)

1 .....

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2 .....

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(ii) The mean rate of urine production on day 1 is 276 cm^3 per hour.

Calculate the mean rate of urine production on day 2 during the 4 hour time period.

(2)

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mean rate of urine production on day 2 = ... cm^3 per hour

(iii) Explain the difference between the mean rate on day 1 and the mean rate on day 2.

(3)

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(Total for Question 1 = 16 marks)

8. Question 1 · part b.iii · 3 marks · Working and answer

Working and answer

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9. Question 6 · part e · 3 marks

November 2020 Paper 2

Further practice · Attempt part e only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

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A person with a mass of 60 kg produces 60 cm^3 of urine an hour and a person with a mass of 80 kg produces 80 cm^3 an hour and so on. Therefore if a 60 kg person drinks 200 cm^3 of beer in an hour they will produce 60 cm^3 of urine plus 320 cm^3, making a total of 380 cm^3.

Alcohol interferes with the mechanism that regulates water levels in our body because alcohol affects a small gland called the pituitary gland. One hormone released by the pituitary gland is anti-diuretic hormone (ADH). Alcohol reduces how much ADH is released by the pituitary. Each unit of alcohol that a person drinks forces the kidneys to produce an extra 120 cm^3 of urine more than the normal output.

Logic suggests that drinking lots of water would replace the 120 cm^3 of urine produced, but it is not that simple. The body only retains half the water that a person drinks. The other half leaves the body in urine, so the person will become dehydrated even if they drink lots of water.

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9. Question 6 · part e · 3 marks

November 2020 Paper 2

Further practice · Attempt part e only.

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(a) (i) Where in the body is the pituitary gland (line 16)?

(1)

- ☒ **A** brain
- ☒ **B** kidney
- ☒ **C** liver
- ☒ **D** pancreas

(ii) Which system includes the pituitary gland (line 16)?

(1)

- ☒ **A** digestive
- ☒ **B** endocrine
- ☒ **C** nervous
- ☒ **D** skeletal

(b) A person weighs 75 kg and drinks 400 cm^3 of beer in an hour (lines 11–14).

(i) Calculate the volume of urine produced by the person in this hour.

(3)

volume of urine = ... cm^3

(ii) Calculate the percentage increase in the volume of urine produced by this person in this hour.

(3)

percentage increase = ...%

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9. Question 6 · part e · 3 marks

November 2020 Paper 2

Further practice · Attempt part e only.

(c) Describe the effect of ADH on the kidney (line 17).

(2)

(d) Suggest the effects of dehydration on the blood after increased urination.

(2)

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(e) Explain why half of the extra water that the person drinks still leaves the body in the urine (lines 21–24).

(3)

(Total for Question 6 = 15 marks)

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9. Question 6 · part e · 3 marks · Working and answer

Working and answer

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